

2. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int source, destination;
    char buffer[1024];
    ssize_t bytes;

    // Open source file
    source = open("source.txt", O_RDONLY);

    if (source < 0)
    {
        perror("Error opening source file");
        return 1;
    }

    // Open destination file
    destination = open("destination.txt",
                      O_WRONLY | O_CREAT | O_TRUNC,
                      0644);

    if (destination < 0)
    {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    // Read from source and write to destination
    while ((bytes = read(source, buffer, sizeof(buffer))) > 0)
    {
        write(destination, buffer, bytes);
    }

    // Close both files
    close(source);
    close(destination);

    printf("File copied successfully.\n");

    return 0;
}
```

```
venkatesh@LAPTOP-LL9RQ023:~$ nano week2.c
venkatesh@LAPTOP-LL9RQ023:~$ cat week2.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int source, destination;
    char buffer[1024];
    ssize_t bytes;

    // Open source file
    source = open("source.txt", O_RDONLY);

    if (source < 0)
    {
        perror("Error opening source file");
        return 1;
    }

    // Open destination file
    destination = open("destination.txt",
                      O_WRONLY | O_CREAT | O_TRUNC,
                      0644);

    if (destination < 0)
    {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    // Read from source and write to destination
    while ((bytes = read(source, buffer, sizeof(buffer))) > 0)
    {
        write(destination, buffer, bytes);
    }

    // Close both files
```

```
int main()
{
    int source, destination;
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    // Open source file
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    // Open destination file
    destination = open("destination.txt",
                      O_WRONLY | O_CREAT | O_TRUNC,
                      0644);

    if (destination < 0)
    {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    // Read from source and write to destination
    while ((bytes = read(source, buffer, sizeof(buffer)))
    {
        write(destination, buffer, bytes);
    }

    // Close both files
    close(source);
    close(destination);

    printf("File copied successfully.\n");

    return 0;
}
```

```
venkatesh@LAPTOP-LL9RQ023:~$ echo "hrlllo this source">source.txt
venkatesh@LAPTOP-LL9RQ023:~$ gcc week2.c
venkatesh@LAPTOP-LL9RQ023:~$ ./a.out
File copied successfully.
venkatesh@LAPTOP-LL9RQ023:~$ cat copy.txt
cat: copy.txt: No such file or directory
venkatesh@LAPTOP-LL9RQ023:~$ cat destination.txt
hrlllo this source
venkatesh@LAPTOP-LL9RQ023:~$ |
```